

SE Sem-IV

MDM (Fundamentals of Telecommunication Systems)

IA-1 Question Bank

Module 1

- State and define the four fundamental characteristics of a data communications system.
- Discuss importance and applications of telecommunication.
- Discuss different forms of data representation (text, number, image, audio, video).
- State key features of different generations(1G-5G) of telecommunication technology.
- Explain different modes of data flow with an example each.
- Discuss evolution of telecommunication.

Module 2

- Explain with example analog and digital signal.
- State the different radio and microwave frequency bands with their related telecommunication applications and frequency range.
- Explain the following transmission media in detail with the necessary diagrams:
 - Twisted pair wire
 - Coaxial cable
 - Fibre optic cable
 - RF
 - Microwave
 - Infrared

Module 3

- Draw the diagram, explain and give advantages and disadvantages of the following network topologies:
 - Mesh
 - Star
 - Bus
 - Ring
 - Hybrid
- Explain the following connecting devices:
Hub, Bridge, Switch, Router.
- Draw the diagram and explain the following Switching techniques:
 - Circuit Switching
 - Packet Switching
- Give an overview of Routing